Problem Set 2

Spring 2024. CSE 309: Theory of Automata

Instructions: This homework is due on **Friday, May 10, 2024**. You are welcome to collaborate on problem sets, provided that (1) you write up your solutions individually, and (2) you clearly cite the names of all collaborators and sources. Failure to do so will result in *zero credit*. An additional key requirement is that you should be able to explain what you submit. Inability to do so will also result again in zero credit. *Direct your all queries to the instructors.*

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1. Prove that the class of *context-free languages* is not closed under either intersection or complementation. [**Hint:** the class of context-free languages is known to be closed under union.]

2. Show that, *every infinite Turing-recognizable language has an infinite Turing-decidable subset.*

3. Let $\text{ALL}_{\text{DFA}} = \{\langle A \rangle : A \text{ is a DFA and } L(A) = \Sigma^*\}$. Prove that ALL_{DFA} is decidable.

4. Let $T = \{\mathtt{a}^i\mathtt{b}^j\mathtt{c}^k : i, j, k \in \mathbb{N}\}$. Prove that T is *countable*, here $\mathtt{a}, \mathtt{b}$, $\mathtt{c}$ are symbols from some alphabet.

5. If L_1 and L_2 are two Turing recognizable languages, show that their concatenation is also Turing recognizable. To do this, assume that you have Turing machines recognizing L_1 and L_2. Then explain how to construct a Turing machine that recognizes L_1L_2. Describe your construction in English. Feel free to give your Turing machine extra working tapes (if needed).
